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The future sustainability of Jute and Tea production in Bangladesh under changing climate conditions using Machine Learning Models

Maruf Hasnain[1], Jannatul Nayeem[2], Mahmudul Hasan Jobaier[3], Md. Ishraq Uddin Sikder[4], Amad Uddin Osama[5]
[1,2,3,4,5,6,7]Department of Computer Science and Engineering, International Islamic University Chittagong
Kumira, Chattogram-4318, Bangladesh
Email- [1]in2farib@gmail.com, [2]jannatul.nayeem6053@gmail.com, [3] jobaieer.cse@gmail.com, [4]nirzon.sikderbd@gmail.com, [5]amaduddinos

***Abstract*—Climate change is expected to impact crop yields globally, especially for economically important crops like tea and jute. This study combines historical yield and climate data with future temperature projections (SSP5-8.5) to forecast yield changes for 36 tea-producing and 20 jute-producing countries by 2050. Using machine learning, Random Forest was the best model to predict tea yields (R^2 = 0.9434), and K-Nearest Neighbors performed best for jute (R^2 = 0.9694). Projections show a decline of around 13.35% in tea yields, while jute yields remain mostly stable due to their broader temperature tolerance in Bangladesh. However, future risks from rainfall changes and water availability may still affect jute. These findings underline the need for crop-specific climate adaptation strategies to support agricultural resilience in a warming world.**

***Index Terms*—Climate change, Crop yield prediction, Tea, Jute, Machine learning, Random Forest, K-Nearest Neighbors (KNN), SSP5-8.5 scenario**

I. Introduction

Climate change is rapidly transforming global agriculture. Rising temperatures, shifting rainfall patterns, and more frequent extreme weather events are already impacting crop productivity and putting the future of vital agricultural systems at risk [1], [2].

Jute, often referred to as 'golden fiber' because of its color and economic value, supports the livelihoods of over 3.5 million farmers in South Asia alone [3]. As a fully biodegradable and renewable fiber, jute presents an eco-friendly alternative to plastic, especially in packaging and textiles, where plastic pollution continues to grow as a global concern. Tea, meanwhile, is one of the world's most widely consumed beverages and a major export commodity. It is primarily grown in hilly regions, providing employment to millions, particularly women, and making significant contributions to national economies [4]. Together, jute and tea not only sustain rural communities but also contribute toward broader sustainable development goals.

Despite their importance, both crops are facing serious threats from ongoing climate change. The Intergovernmental Panel on Climate Change [1] forecasts a global temperature rise of 1.5 to 2 °C by 2050, with even more severe warming under high-emission scenarios. For jute, higher temperatures, erratic monsoons, and prolonged flooding weaken fiber strength and quality, while drought conditions stunt growth. Tea cultivation is similarly affected, with heat stress impacting leaf development and increased humidity encouraging pest and disease outbreaks. Real-world examples already reflect these risks: Assam's jute farmers have been hit hard by monsoon floods [5], and in Bangladesh's Gaibandha district, rising temperatures and reduced rainfall have degraded fiber quality [6]. Studies predict that a 2 °C temperature increase in tea could significantly reduce yield and export quality by mid-century [7].

Previous research confirms these vulnerabilities. Parry et al. [8] highlighted how rising temperatures threaten crop yields in sensitive regions like South Asia. Islam, Md Begum [9] documented how erratic monsoons disrupt jute farming in Bangladesh. Similarly, Wijeratne et al. [10] found that fluctuations in temperature and rainfall reduce tea yield and quality in Sri Lanka. These studies emphasize the urgent need to develop adaptation strategies that can help safeguard these crops and the communities that depend on them.

Building on this foundation, our paper aims to assess how climate change is impacting jute and tea production across major producing countries and to explore what the future may hold under significant warming scenarios. Using historical data from FAOSTAT and WorldClim, combined with future climate projections, we seek to quantify potential impacts and highlight the pressing need for targeted interventions.

The specific objectives of this study are to:

- Forecast the expected changes in jute and tea yields by 2050 under the SSP5-8.5 climate scenario for key producing countries.
- Identify and propose practical adaptation and resilience strategies to help jute and tea farming systems cope with

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projected climate impacts and protect rural livelihoods.

The paper is structured as follows: Section 2 reviews previous research on the climate-related challenges facing jute and tea. Section 3 describes the proposed Methodology. Section 4 presents the results on potential impacts to yield, quality, and livelihoods. Finally, Section 5 concludes with key recommendations for building resilience in jute and tea cultivation under a changing climate.

II. Related Work

Global agriculture faces unprecedented challenges from climate change, necessitating robust predictive tools to understand future impacts on crop production. Rezaei et al. [11] conducted an extensive review of climate change impacts on global crop yields, focusing on staple cereals such as wheat, maize, rice, millet, and sorghum. Their approach integrated results from multifactorial field experiments and process-based modeling to assess the influence of elevated CO, rising temperatures, and altered precipitation patterns. The study highlighted that while elevated CO can boost yields in C3 crops like wheat and rice, these gains are often offset by heat and drought stresses—especially in C4 crops such as maize. Adaptation strategies, including irrigation and nutrient optimization, were shown to mitigate yield losses, particularly in higher latitudes. The authors emphasized the need for enhanced experimental designs and data-driven models to improve yield predictions under climate change conditions.

The advent of advanced computational methods has seen a surge in the application of machine learning (ML) for agricultural forecasting. Kheir et al. [12] developed an automated machine learning (AutoML) approach for fast and robust prediction of wheat yields in Egypt. By integrating extensive field survey yield data with a diverse fusion of remote sensing (vegetative indices), soil, and weather datasets, their study utilized the H2O AutoML framework to train and evaluate 22 ML models. A stacked ensemble (super learner) demonstrated high accuracy ($d = 0.82$, $R^2 = 0.51$). The trained models were successfully deployed to predict the yield using only remote sensing data ($R^2 = 0.7$) for the initial possible estimation and under the climate change scenarios of CMIP6, projecting declines in wheat yield of 5% (SSP2-4.5) and 21% (SSP5-8.5) in the middle of the century. Their work offers a rapid, accurate, and low-cost method for yield forecasting, especially in data-scarce regions, aiding stakeholders in decision-making processes.

Hybrid modeling, which combines the strengths of mechanistic crop models and data-driven ML techniques, is also gaining traction. Qing Sun et al. [13] proposed such a framework, integrating Random Forest (RF) machine learning models with nine global gridded crop models (GGCMs) to enhance the accuracy of maize and soybean yield projections under future climate scenarios. By incorporating environmental variables such as chilling days, drought, and crop pests and diseases, the GGCM+RF models demonstrated significant improvements in predictive performance, increasing correlation coefficients (r) and reducing normalized root mean square error (nRMSE) compared to GGCMs alone. This approach effectively constrained uncertainties in yield projections by 33–78% for maize and 56-68% for soybeans, offering a robust framework for assessing the impacts of climate change on crop yields.

Further exploring the integration of crop models and machine learning, Zhang et al. [14] developed an integrated framework combining the APSIM crop model with Random Forest algorithms to optimize maize hybrid selection under future climate scenarios in Northeast China. Their study assessed the performance of 54 maize hybrids, successfully identifying optimal hybrid-environment combinations and demonstrating that integrating crop models with machine learning can effectively guide cultivar selection and adaptation strategies.

Specific regional applications of ML further illustrate its utility. Dhillon et al. [15] utilized a machine learning framework to evaluate the effect of climate change on maize and soybean yields in Ohio, USA. By employing Random Forest algorithms with historical monthly county-level weather data (temperature and precipitation), their study quantified temporal yield variability and projected yield changes for both crops up to the year 2100 under different climate emission scenarios. The approach predicted significant yield reductions (for example, up to 18. 5% for maize and 9. 63% for soybean with higher emissions) and identified key influential weather parameters, demonstrating the ability of machine learning to model complex climate-yield relationships. Their analysis also highlighted substantial economic implications, with losses projected between 254.7 million dollar and 751.1 million dollar for maize and soybeans, respectively, under lower emission scenarios, emphasizing the need for such predictive models to inform mitigation and adaptation efforts.

The reviewed literature clearly demonstrates the profound impacts of climate change on global agricultural yields, particularly for staple cereals, and underscores the growing efficacy of machine learning, automated ML, and hybrid modeling approaches in forecasting these impacts. These studies effectively integrate diverse datasets (remote sensing, soil, weather) and project future scenarios, often focusing on major commodity crops in various regions. However, dedicated and detailed forecasting using these advanced predictive frameworks for economically and ecologically vital, yet often less-studied, crops like jute and tea, especially within the climate-vulnerable agricultural landscapes of South Asia, represents a critical area for further investigation. This present study aims to address this gap by adapting and applying robust predictive modeling techniques to specifically forecast the future changes in yield, quality, and optimal cultivation zones for jute and tea under relevant climate change scenarios. The ultimate goal is to furnish actionable insights that can guide the development of targeted adaptation strategies, thereby enhancing the resilience of these key agricultural sectors and the communities that depend on them.

III. PROPOSED METHODOLOGY

A. Problem Statement

Climate change poses a significant and escalating threat to the production of jute and tea, crops vital to the economies and livelihoods of numerous countries, particularly within South Asia (e.g., India, Bangladesh for jute and tea; Sri Lanka for tea) but also encompassing other key producers across Asia (e.g., China, Viet Nam, Indonesia for both) and Africa (e.g., Kenya, Tanzania, Malawi for tea), as indicated by the scope of available data. The core problem is the current deficiency in robust, quantitative forecasts that detail the specific future impacts of climate change, such as altered temperatures and rainfall patterns - on the yield, quality, and geographic suitability for these two distinct crops within these diverse producing nations. This lack of granular, crop-specific predictive insight, especially when compared to more extensively studied staple cereals, impedes the formulation and implementation of effective, regionally-tailored adaptation and resilience-building strategies. Consequently, this gap jeopardizes the long-term sustainability of these critical agricultural sectors and the food security and economic stability of the communities reliant upon them across these specific geographies.

B. About Dataset

This study utilizes several key datasets to analyze the historical and projected impacts of climate change on jute and tea production. These include historical crop production records, historical climate observations, and future climate projections.

C. About Data

For historical analysis spanning 1961 to 2021, this study used crop-specific data streams covering production, temperature, and rainfall. Annual production data (specify units here, e.g., "in tonnes") for both jute and tea were sourced from the Food and Agriculture Organization Corporate Statistical Database (FAOSTAT). This dataset provided records for a comprehensive list of producing countries. For jute, data for: Bangladesh, Bhutan, Brazil, Cambodia, Cameroon, China, Egypt, El Salvador, India, Iran (Islamic Republic of), Japan, Myanmar, Nepal, Pakistan, Peru, South Sudan, Thailand, Uzbekistan, Viet Nam, and Zimbabwe. For tea, countries included: Argentina, Azerbaijan, Bangladesh, Bolivia (Plurinational State of), Brazil, Burundi, Cameroon, China, Colombia, Ecuador, El Salvador, Ethiopia, Georgia, Guatemala, India, Indonesia, Japan, Kenya, Madagascar, Malawi, Malaysia, Mali, Mauritius, Montenegro, Mozambique, Nepal, Panama, Papua New Guinea, Peru, Portugal, Rwanda, Seychelles, South Africa, Sri Lanka, Thailand, Uganda, United Republic of Tanzania, Viet Nam, Zambia, and Zimbabwe. Alongside this production data, corresponding historical mean temperature and total precipitation records for these nations and years were integrated to assess climate influences during this period.

Temperature data, integral to assessing climatic influences, were sourced for both the historical period corresponding to the crop data (1961-2021) and for future projections (2021-2050). Historical monthly mean temperature data (in °C) for the 1961-2021 timeframe were primarily obtained from the WorldClim dataset , a gridded global dataset offering high-resolution climate surfaces relevant to the geographical locations of the jute and tea-producing countries under study. For future temperature conditions extending from 2021 to 2050, projections were based on outputs from General Circulation Models (GCMs) from the Coupled Model Intercomparison Project Phase 6 (CMIP6). Specifically, this study utilized the Shared Socioeconomic Pathway SSP5-8.5. The SSP5-8.5 scenario depicts a high-emission, fossil-fuel-intensive development pathway, representing a future with substantial global warming, and was employed to forecast potential temperature changes and their consequent impacts. These future projections, often accessible via resources like WorldClim which provide downscaled CMIP6 outputs, supplied the projected monthly mean temperature data necessary for evaluating the potential impacts on jute and tea production under a significant future warming trajectory.

D. Data Preprocessing

In this study, we used data on crop production, past climate records, and future climate predictions. These were cleaned and prepared to make them ready for analysis and modeling.

First, we collected yearly production and yield data for jute and tea from 1961 to 2021 using the FAOSTAT database. We made sure the units were the same across all years and countries, such as tonnes per hectare for yield and total tonnes for production.

Next, we got past climate data from WorldClim, including average monthly temperature and total rainfall. These were also for the years 1961 to 2021. Since the climate data came in grids, we adjusted them to match each country's level to fit with the crop data. If any values were missing, we filled them in by estimating where possible. When that wasn't possible, we left out those country and year combinations.

Then, we added future climate data from CMIP6 models using the SSP5-8.5 scenario. These were available through sources like WorldClim and covered the years from 2021 to 2050. We formatted them to look like the past climate data so they could be used in the same way.

At the end, we combined all the data - production, past climate, and future climate - into a single, organized file. This was used for exploring the data and for training the machine learning models to predict how climate change might affect jute and tea crops.

E. Data Exploration and Feature Extraction

To prepare for predicting climate change impacts, we first closely examined our historical data for jute and tea production (from 1961 to 2021) alongside temperature and rainfall records for the same period. This involved looking at basic patterns, like how production and climate variables trended over the years, and how they might be related to each other – for example, if crop yields were often higher or lower in years

with certain temperature or rainfall conditions. From this understanding, the key pieces of information (features) selected to feed into our predictive models were the annual production figures for jute and tea, along with historical monthly temperature and rainfall data for each producing country. We also included the 'Year' as a feature to help account for any ongoing trends not directly related to climate, such as technological advancements. For forecasting future impacts, projected temperature data under the SSP5-8.5 scenario (and corresponding rainfall data) from 2021 to 2050 were then prepared in a consistent format to match the historical data our models were trained on.

F. Model Comparison and Selection

To understand how climate change might affect jute and tea production in the future, we trained and tested a few machine learning models using the cleaned historical data. For both crops, we used models like Linear Regression, Decision Tree, K-Nearest Neighbors (KNN), and Random Forest. We checked how well each model performed by looking at the R^2 score, which shows how accurate the predictions were. This was done using a separate test set (or cross-validation, depending on the case). The goal was to find the most reliable model for each crop so we could use it to make future predictions.

1) Linear Regression: Linear Regression is a statistical method used to model the linear relationship between a dependent variable and one or more independent variables [16].

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n + \epsilon \tag{1}$$

where

- Y is the predicted crop yield (jute or tea),
- β_0 is the intercept,
- $\beta_1, \beta_2, \ldots, \beta_n$ are the coefficients for climate variables (e.g., temperature, rainfall),
- $X_1, X_2, \ldots, X_n$ are the independent climate features,
- ϵ is the error term.

This model served as a baseline to examine whether simple linear relationships existed between climate parameters (temperature and rainfall) and crop yield. It provided interpretable results and showed which variables had the most positive or negative effect on production. However, it was limited in capturing complex, non-linear patterns in the data.

2) Decision Tree: A Decision Tree is a supervised learning algorithm that splits data into branches based on decision rules to make predictions [17] It recursively divides the dataset into subsets based on feature thresholds that maximize *information gain* or minimize *mean squared error (MSE)* for regression tasks.

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y})^2 \tag{2}$$

The Decision Tree model helped identify threshold-based effects of climate variables on yields, such as if average temperature is greater than 28°C, then tea yield decreases. It handled non-linear relationships and interactions well but was prone to overfitting on small datasets.

3) K-Nearest Neighbors (KNN): KNN is a non-parametric method that predicts the output for a given input based on the outputs of its k-nearest neighbors in the feature space [18] .

$$d(x, x_i) = \sqrt{\sum_{j=1}^{n}(x_j - x_{ij})^2} \tag{3}$$

where

- x is the input feature vector,
- x_i is the feature vector of the i-th neighbor,
- d is the distance metric (Euclidean).

Prediction:

$$\hat{y} = \frac{1}{k}\sum_{i=1}^{k} y_i \tag{4}$$

KNN was applied to identify local trends in crop yield patterns. It was especially useful for countries with highly irregular climate conditions. By relying on similar past years (neighbors), it could make robust predictions without assuming a functional form of the data.

4) Random Forest: Random Forest is an ensemble method that combines multiple decision trees trained on different subsets of the data and features to improve prediction accuracy and reduce overfitting.

Working Principle:

- Build multiple decision trees using *bootstrap samples* of the dataset.
- For each split in the tree, select the best feature from a random subset of features.
- Final prediction is the *average of all tree predictions* (for regression).

$$\hat{y} = \frac{1}{T}\sum_{t=1}^{T}\hat{y}^{(t)} \tag{5}$$

where

- T is the number of trees,
- $\hat{y}^{(t)}$ is the prediction from the t-th tree.

Random Forest delivered the best predictive performance in most cases. It handled high-dimensional data well and modeled complex interactions between climate variables and crop yields. It also provided *feature importance* rankings, aiding in feature selection. This model was ultimately used for final forecasting of jute and tea production under the SSP5-8.5 climate scenario.

IV. Results and Discussion

In this section, we evaluate how effectively our machine learning models predicted historical tea and jute yields across multiple countries and examine what they reveal about the future of these crops under a high-emissions climate scenario (SSP5-8.5).

To identify the most reliable forecasting methods, we tested several machine learning models using historical climate data

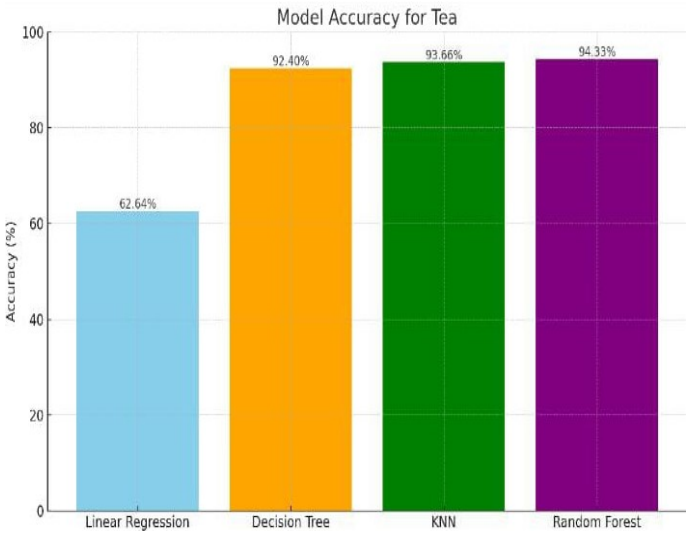

Fig. 1: Models Accuracy for Tea

and yield records from major tea- and jute-producing countries. We assessed model accuracy using R^2 scores, which measure how closely the model's predictions match actual outcomes—a score closer to 1 indicates a better fit. The performance results are summarized in Figure 1 for tea and Figure 2 for jute.

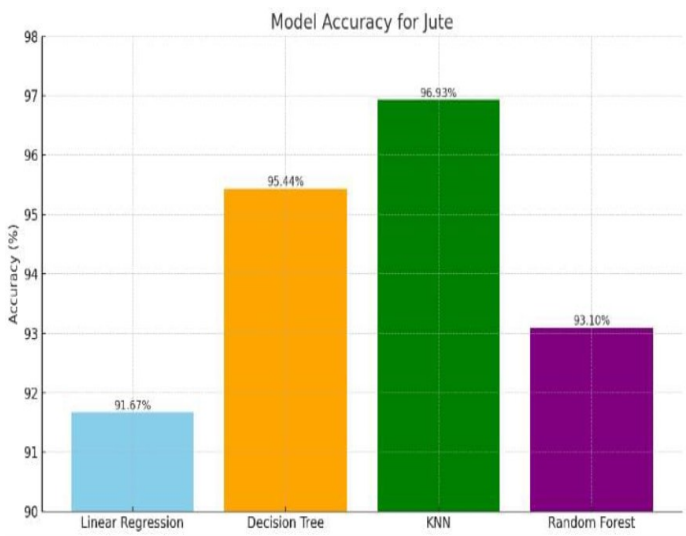

Fig. 2: Models Accuracy for jute

For tea, which included data from 36 countries, the Random Forest model was the most accurate, achieving an R^2 score of 0.9434 (94.34%). This high score suggests the model captured complex climate patterns and their effects on tea yields very effectively, as seen in Figure 1. In the case of jute, based on data from 20 countries or regions, the K-Nearest Neighbors (KNN) model delivered the best performance, with an R^2 score of 0.9694 (96.94%). Figure 2 shows how KNN outperformed the other models, indicating that jute yields are strongly influenced by conditions similar to past ones, which this model handles well.Given these strong results, we used the Random Forest model to forecast future tea yields and the KNN model for future jute yields.

We then applied these models to estimate how tea and jute production might change from 2021 to 2050 under the SSP5-8.5 scenario - a pathway that assumes high greenhouse gas emissions and significant global warming.

Our projections show that tea yields are likely to decline significantly under SSP5-8.5. In Bangladesh, the average tea yield is expected to decrease by approximately 12–15% by 2050 .

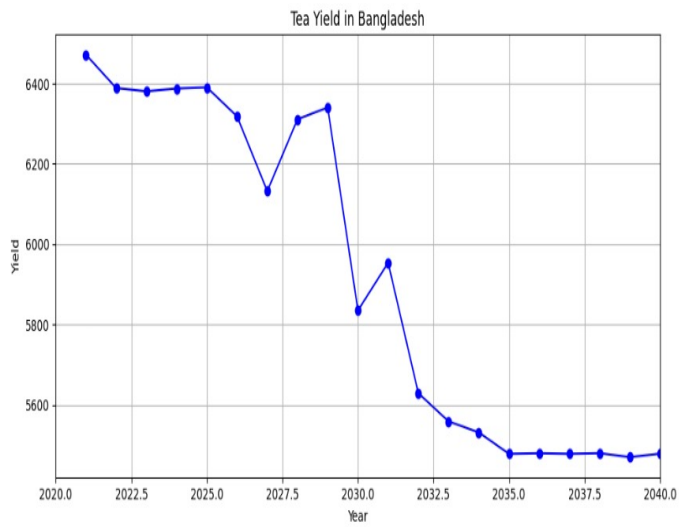

Fig. 3: Tea Prediction

As a case in point, Figure 3 shows the forecast for Bangladesh. Initially, tea yields remain relatively stable through the 2020s, with minor fluctuations. However, starting around 2030, yields begin a sharp and sustained decline, eventually stabilizing at a much lower level-around 5500 kg/ha, compared to roughly 6450 kg/ha in the early 2020s. This represents a 13.35% decrease in productivity for Bangladesh by 2040.

This pattern of initial stability followed by decline appears in several other tea-producing countries as well. It suggests that critical environmental thresholds - such as tolerance to rising temperatures or changes in seasonality - may be crossed around 2030, triggering a drop in productivity.This decline is especially concerning because tea is known to be sensitive to heat stress. High temperatures can reduce leaf quality, shorten the growing period, and increase the risk of pests and diseases (Wijeratne et al., 2007; Climate Fact Checks, 2023). The drop in yields would affect not only farmers' incomes, especially smallholders, but also national export earnings in several tea-dependent countries.

In contrast to tea, the outlook for jute is relatively more stable. When we ran the KNN model using projected temperature increases (about +2°C by 2050 in many of the producing regions), the model did not show any significant change in average jute yield. This resilience can be explained by jute's

wide optimal temperature range, typically between 25°C and 37°C. In many regions, the projected temperature rise doesn't push growing conditions beyond this threshold, especially if current temperatures aren't already near the upper limit. This gives jute a thermal buffer that tea does not seem to have.

However, it's important to point out that our projections focused only on temperature. Jute is highly dependent on other climate factors, especially rainfall. Water is critical for the retting process (a key stage in jute fiber preparation), and shifts in rainfall patterns, monsoon timing, or water availability could still pose serious risks (Rahman et al., 2017; STM Journals, 2022).So, while jute may appear stable under rising temperatures, this should not be mistaken for overall climate resilience. Future projections must also account for changing precipitation, droughts, and water stress—factors not captured in the current model run. The results show two very different climate futures for tea and jute. Tea is clearly vulnerable to temperature rises, with many countries likely to see double-digit declines in yield by 2050. Jute, while more resilient to heat, remains susceptible to water-related stresses that were not fully explored in this study.

These findings underscore the need for tailored climate adaptation strategies. For tea, urgent steps may include introducing heat-tolerant varieties, shade management, and early warning systems for pests and diseases. For jute, the focus may need to be on irrigation, retting infrastructure, and rainwater harvesting.

Our predictions are based on one climate scenario (SSP5-8.5) and focus mainly on temperature, with limited consideration of factors like CO effects, extreme events, or local adaptation. Results are at the national scale, so regional variations may not be captured.Future work should incorporate a broader range of climate scenarios (including explicit rainfall projections for both crops), integrate CO effects, and model at finer spatial resolutions. Investigating the combined effects of temperature and altered water availability on jute, and exploring effective adaptation options for both crops, are critical next steps.

V. Conclusions

This study highlights the contrasting climate resilience of tea and jute under a high-emission future. Tea yields are projected to decline significantly as temperatures rise, showing the crop's sensitivity to heat stress. In contrast, jute yields are expected to remain stable, indicating greater tolerance to projected temperature increases. These findings emphasize the need for targeted adaptation strategies. For tea, efforts should focus on developing heat-resilient practices, while for jute, ensuring reliable water availability and managing rainfall variability will be key to sustaining production and protecting farmer livelihoods in the years ahead.

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